June 17, 2026

June 14, 2026

Delft University of Technology, Delft, Netherlands.

Dear Dr. Jing Sun,

I am excited to apply for the fully funded PhD position in Scientific Machine Learning (SciML) at Delft University of Technology, as advertised on [Original Link](https://www.phdscanner.com/opportunities/phd-vacancies-delft-university-of-technology-netherlands-phd-position-scientific-machine-learning-toward-scientific-foundation-models-a42e35d2-c94d-409b-804a-b8411439be5f). As a highly motivated and detail-oriented AI Data Science Engineer, I am confident that my skills and experience make me an ideal candidate for this project.

My research background is deeply rooted in SciML, with a strong focus on developing foundation models for complex systems modeling. During my Master's thesis, I worked on the "Cardiovascular Medical Digital Twin" project, where I developed several innovative solutions that perfectly align with the requirements of this PhD position. Specifically, I would like to highlight two achievements that demonstrate my expertise in SciML.

Firstly, I developed a linear-time state space model called Medical$_{M}ambaTS, whichenablessub-$
$millisecondbedsideinferenceforcriticalapplicationssuchascardiovasculardiseasediagnosisandtreatment$

Secondly, I designed a 0D Windkessel Physics-Informed Operator Network called PI$_{D}eepONetusingPyTorchAutograd, whichcombinesphysicalconstraintswithmachinelearningtechnique$
$informedneuralnetworksandhybridphysics-MLapproaches, bothofwhichareessentialfordevelopingfou$
$drivenmachinelearningtechniques.$

In addition to these achievements, I have also worked on missingness-aware gate design for handling sensor dropouts and missing data in real-world applications. This experience has taught me the importance of robustness and adaptability in SciML models, which is a key aspect of this PhD project.

I am confident that my skills and experience make me an excellent fit for this PhD position. I am excited about the opportunity to contribute to the development of foundation models for complex systems modeling and explore modern SciML methods such as physics-informed neural networks and hybrid physics-ML approaches.

Thank you for considering my application. I look forward to discussing my qualifications further.

Sincerely,

Ismail Aissaoui